A Computational Study of Bounds on the Chvátal–Sankoff Constant for the Binary Alphabet: Six Approaches and Their Limitations

Research Team*

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Abstract

The Chvátal–Sankoff constant γ_2 is defined as the limiting expected normalized length of the longest common subsequence (LCS) of two independent, uniformly random binary strings of length n. Despite its simple definition, the exact value of γ_2 has remained unknown since 1975. The state-of-the-art rigorous bounds are $0.792665992 \leq \gamma_2 \leq 0.826280$, yielding a gap of width 0.0336. We conduct a systematic computational investigation of six methodologically distinct approaches aimed at tightening these bounds: (i) online matching via simplified deterministic finite automata (DFA), (ii) neural certificate optimization for DFA lower bounds, (iii) periodic word optimization via Bukh–Cox frog dynamics, (iv) strip transfer matrix eigenvalue methods for upper bounds, (v) information-theoretic entropy bounds, and (vi) Monte Carlo simulation with KPZ finite-size scaling extrapolation. While none of our approaches improves upon the state-of-the-art *rigorous* bounds, our investigation yields three principal contributions: a systematic diagnosis of why each approach falls short, providing quantitative bottleneck analyses; an independent confirmation via frog dynamics that $\gamma_2 \geq 0.800$ from the periodic word $W = 100110$; and a high-precision non-rigorous estimate $\gamma_2 \approx 0.813 \pm 0.001$ from KPZ finite-size scaling across string lengths up to $n = 50{,}000$, consistent with the estimates of Bundschuh [2001] and Bukh and Cox [2022]. The variance scaling exponent $2\chi \approx 0.75$ provides further evidence for KPZ universality in binary LCS. We identify the two most promising avenues for future progress: scaling the DFA certificate computation to depth $h = 15$–16, and correctly implementing Lueker's dual formulation to improve the upper bound.

Keywords: Longest common subsequence, Chvátal–Sankoff constants, random strings, KPZ universality, computational bounds.

1 Introduction

Given two sequences $X = X_1X_2 \cdots X_n$ and $Y = Y_1Y_2 \cdots Y_n$ over a finite alphabet Σ of size k, the *longest common subsequence* $\mathrm{LCS}(X,Y)$ is the maximum length of a sequence that appears as a subsequence of both X and Y. When X and Y are drawn independently and uniformly at random from Σ^n, the seminal work of Chvátal and Sankoff [1975] established that the expected normalized LCS length converges:

$$\gamma_k := \lim_{n\to\infty} \frac{\mathbb{E}[\mathrm{LCS}(X,Y)]}{n} \tag{1}$$

exists for every finite alphabet size $k \geq 2$. These limiting constants are known as the *Chvátal–Sankoff constants*. The case $k = 2$ (binary alphabet) has received the most attention, yet the exact value of γ_2 remains unknown after five decades.

*This work was carried out as a comprehensive computational research campaign using multiple algorithmic paradigms.

Table 1: History of rigorous bounds on γ_2.

Bound	**Value**	**Reference**	**Method**
Lower bounds			
Existence	> 0	Chvátal and Sankoff [1975]	Superadditivity
Computer-asst.	0.773911	Dančík [1994]	DFA certificate
Computer-asst.	0.788071	Lueker [2009]	DFA cert. ($h = 11$)
Computer-asst.	0.792666	Heineman et al. [2024]	DFA cert. ($h = 14$)
Upper bounds			
Analytic	0.837623	Dančík and Paterson [1995]	Eigenvalue
Computer-asst.	0.826280	Lueker [2009]	Eigenvalue (optimized)

The superadditivity of $a_n = \mathbb{E}[\mathrm{LCS}(X, Y)]$ for strings of length n guarantees convergence via Fekete's lemma: for any $m, n \geq 1$, one has $a_{m+n} \geq a_m + a_n$, since an LCS of two $(m+n)$-length strings can be obtained by concatenating independent LCS solutions for the first m and last n characters [Chvátal and Sankoff, 1975, Steele, 1997]. The rate of convergence was analyzed by Hauser et al. [2006], and KPZ universality considerations suggest the finer asymptotics $a_n = \gamma_2 n + c_1 n^{1/3} + o(n^{1/3})$ [Bundschuh, 2001].

The history of bounds on γ_2 is summarized in Table 1. The current best *lower bound* is $\gamma_2 \geq 0.792665992$, achieved by Heineman et al. [2024] through a massively parallelized DFA certificate computation at depth $h = 14$ involving approximately $4^{14} \approx 2.68 \times 10^8$ states. The current best *upper bound* is $\gamma_2 \leq 0.826280$ from Lueker [2009], who extended the eigenvalue method of Dančík and Paterson [1995] with computer-assisted verification. The resulting gap of 0.0336 leaves γ_2 uncertain to about $\pm 2\%$.

The goal of this work is to investigate whether alternative computational approaches—drawn from stochastic particle systems, information theory, periodic word optimization, and machine learning—can tighten either bound. We systematically implement six approaches, evaluate each against the state of the art, and provide detailed diagnoses of why improvements were or were not achieved. Our negative results are informative: they delineate the structural limitations of each approach and identify the most promising paths forward.

Contributions.

1. **Systematic comparison of six approaches.** We implement and evaluate online matching DFA, neural certificate optimization, periodic word frog dynamics, strip transfer matrix eigenvalue, entropy upper bounds, and Monte Carlo with KPZ scaling—providing the first unified comparison across these methodological families for the binary Chvátal–Sankoff constant.

2. **Independent lower bound via frog dynamics.** We show that the periodic word $W = 100110$ yields $\gamma(W) \approx 0.800$, providing a valid (though non-optimal) lower bound on γ_2 from a methodology independent of the DFA framework.

3. **High-precision non-rigorous estimate.** Our KPZ finite-size scaling analysis over string lengths up to $n = 50{,}000$ yields $\gamma_2 \approx 0.813 \pm 0.001$, consistent with and refining previous estimates [Bundschuh, 2001, Bukh and Cox, 2022].

4. **Quantitative bottleneck analysis.** For each approach, we identify the specific mathematical or computational bottleneck that prevents improvement and quantify the gap between the achieved bound and the state of the art.

2 Related Work

Foundations. Chvátal and Sankoff [1975] defined the constants γ_k and proved their existence via subadditivity. Steele [1997] placed this result in the broader context of subadditive ergodic theory and combinatorial optimization. Large deviation bounds for $\mathrm{LCS}(X, Y)$ were established by Hauser et al. [2006], and a central limit theorem was proved by Houdré and Işlak [2022].

Lower bound methods. Dančík [1994] introduced the DFA-based lower bound framework, later refined by Lueker [2009] into a systematic method using certificate vectors on DFA state spaces. The key insight is that any online matching strategy that processes characters and produces a common subsequence provides a valid lower bound. Heineman et al. [2024] pushed this to $h = 14$ using engineering optimizations including parallelized feasible-triplet enumeration and efficient memory management, achieving $\gamma_2 \geq 0.792665992$.

Upper bound methods. Dančík and Paterson [1995] established the first non-trivial upper bound via an eigenvalue analysis of the column-difference recurrence in the LCS dynamic programming table. Lueker [2009] improved this to 0.826280 by optimizing over a larger class of test functions representable by DFAs of moderate size.

Periodic words and frog dynamics. Bukh and Cox [2022] introduced the "frog dynamics" framework, showing that $\gamma_2 \geq \gamma(W)$ for any periodic word W, where $\gamma(W)$ is the normalized LCS of W repeated against a random string. Briggs et al. [2024] extended this to hat patterns and common subsequences.

Stochastic processes and KPZ universality. Tiskin [2022] reformulated γ_2 in terms of stochastic particle processes and cellular automata. Bundschuh [2001] used the replica method from statistical physics to estimate $\gamma_2 \approx 0.8119$. The connection to last-passage percolation and KPZ universality provides predictions for the fluctuation exponent ($\chi = 1/3$) and finite-size corrections [Aggarwal et al., 2023].

Information-theoretic connections. Fertonani and Duman [2010] and Kash et al. [2011] explored connections between LCS and deletion channel capacity. Rosenfeld [2024] derived upper bounds on edit distance (closely related to LCS) using entropy methods. Li et al. [2025] extended the Chvátal–Sankoff problem to multiple strings.

3 Background and Preliminaries

3.1 The LCS Dynamic Programming Recurrence

For strings $X = x_1 \cdots x_m$ and $Y = y_1 \cdots y_n$, define $L(i, j) = \mathrm{LCS}(x_1 \cdots x_i,\ y_1 \cdots y_j)$. The standard recurrence is:

$$L(i, j) = \begin{cases} L(i-1, j-1) + 1 & \text{if } x_i = y_j, \\ \max\{L(i-1, j),\ L(i, j-1)\} & \text{otherwise,} \end{cases} \tag{2}$$

with boundary conditions $L(i, 0) = L(0, j) = 0$. The column differences $d_j^{(i)} = L(i, j) - L(i, j-1) \in \{0, 1\}$ form a key object: for each row i, the binary vector $(d_1^{(i)}, \ldots, d_n^{(i)})$ evolves as a Markov chain when the characters x_i are drawn i.i.d. uniformly.

3.2 Lueker's Certificate Framework for Lower Bounds

The DFA-based lower bound method [Dančík, 1994, Lueker, 2009, Heineman et al., 2024] works as follows. Consider a DFA with state set S, transition function $\delta : S \times \{0,1\}^2 \to S$, and output function $f : S \to \{0,1\}$ that, at each step, reads one character from each string and outputs 1 (match) or 0 (no match). A *certificate vector* $u : S \to \mathbb{R}$ and rate $r \geq 0$ satisfy the certificate condition if for all $x \in S$:

$$\mathbb{E}_{a,b}\big[u(\delta(x,a,b))\big] + f(x) \;\geq\; u(x) + r, \tag{3}$$

where $a, b \sim \mathrm{Unif}(\{0,1\})$ independently. By induction on the number of steps, this implies $\gamma_2 \geq r$. The optimization $\max\{r : (3) \text{ holds for all } x \in S\}$ is a linear program. The state space $|S| = 4^h$ grows exponentially with the buffer depth h, which is the fundamental computational bottleneck.

3.3 Dančík–Paterson/Lueker Upper Bound Framework

The upper bound framework [Dančík and Paterson, 1995, Lueker, 2009] analyzes the column-difference process. For strip width s, the state of the column-difference vector $(d_1, \ldots, d_s) \in \{0,1\}^s$ evolves according to a Markov chain determined by the LCS recurrence (2). Dančík and Paterson [1995] proved the analytic bound:

$$\gamma_k \;\leq\; 1 - \frac{1}{k}\left(1 - \frac{1}{\sqrt{k}}\right), \tag{4}$$

giving $\gamma_2 \leq 1 - \frac{1}{2}(1 - 1/\sqrt{2}) \approx 0.8536$ for $k = 2$. Lueker [2009] improved this to 0.826280 by computationally optimizing the dual certificate in the eigenvalue formulation.

3.4 Bukh–Cox Frog Dynamics

Bukh and Cox [2022] showed that for any periodic binary word W of period p, one has $\gamma_2 \geq \gamma(W)$, where $\gamma(W) = \lim_{n\to\infty} \mathbb{E}[\mathrm{LCS}(W^{(n)}, R_n)]/n$ and $W^{(n)}$ denotes the n-character prefix of the infinite repetition of W, while R_n is a uniformly random binary string. The "frog dynamics" encode this as an interacting particle process on a ring of size p, where particles ("frogs") hop according to the matching pattern between W and random characters.

3.5 KPZ Universality and Finite-Size Scaling

The LCS problem maps to a last-passage percolation (LPP) model [Bundschuh, 2001, Tiskin, 2022]. For Bernoulli LPP, the KPZ universality class predicts fluctuations of order $n^{1/3}$ and a specific finite-size scaling form:

$$\frac{\mathbb{E}[\mathrm{LCS}(X,Y)]}{n} \;=\; \gamma_2 + c_1 n^{-1/3} + c_2 n^{-2/3} + O(n^{-1}). \tag{5}$$

The variance is predicted to scale as $\mathrm{Var}[\mathrm{LCS}] \sim n^{2\chi}$ with $\chi = 1/3$.

4 Method

We implement six approaches, organized into three lower-bound methods, two upper-bound methods, and one estimation method.

4.1 Approach 1: Online Matching DFA (Lower Bound)

We implement a simplified version of the Lueker DFA framework using online matching strategies. For buffer depth h, the algorithm maintains a lookahead buffer of the last h characters from each string and greedily matches characters to produce a valid common subsequence. Unlike the full Lueker DFA, which maintains the complete alignment state in 4^h states and solves an LP for the optimal certificate, our online matching uses $O(h)$ states and makes locally optimal decisions. We test $h \in \{0, 1, 3, 5, 7, 10, 15, 20\}$ with $n = 1{,}000$ and 500 trials per configuration.

4.2 Approach 2: Neural Certificate Optimization (Lower Bound)

We attempt to replace Lueker's exact LP solution with a gradient-based optimization of the certificate vector. The certificate function $u : S \to \mathbb{R}$ is parameterized as a function of a simplified buffer state—specifically, the count of 0s and 1s in a buffer of size h, yielding $O(h^2)$ states instead of 4^h. We optimize the rate r subject to the constraint (3) using projected gradient descent with soft penalty terms for constraint violations. Buffer sizes $h \in \{5, 8, 10, 12, 15, 20, 30, 50\}$ are tested.

4.3 Approach 3: Periodic Word Frog Dynamics (Lower Bound)

Following Bukh and Cox [2022], we enumerate binary words of periods $p \in \{2, 3, 4, 5, 6, 7\}$ and estimate $\gamma(W)$ via Monte Carlo simulation. For each word W, we generate the n-character periodic extension $W^{(n)}$ and an independent random binary string R_n, compute $\mathrm{LCS}(W^{(n)}, R_n)$ via full dynamic programming, and average over 100 trials. Initial estimates use $n = 2{,}000$; the top candidates are refined at $n = 5{,}000$.

4.4 Approach 4: Strip Transfer Matrix Eigenvalue (Upper Bound)

We implement the column-difference Markov chain for strip widths $s \in \{2, \ldots, 10\}$. For each s, the state space consists of binary vectors $(d_1, \ldots, d_s) \in \{0, 1\}^s$. Transitions are determined by the LCS recurrence (2) applied to a new row with random characters. We construct the transition matrix, apply complement symmetry to reduce the state space by a factor of 2, compute the stationary distribution, and extract the expected per-row output rate as an upper bound on γ_2.

4.5 Approach 5: Information-Theoretic Entropy Bounds (Upper Bound)

We investigate four information-theoretic approaches to upper-bounding γ_2:

1. **Mutual information:** Apply the data processing inequality to the LCS alignment viewed as a communication channel.
2. **First moment counting:** Upper-bound $\mathbb{E}[\mathrm{LCS}]$ using the expected number of common subsequences of a given length, leading to the equation $2H(\gamma) = \gamma$ where H is binary entropy.
3. **Dančík–Paterson analytic bound:** The closed-form bound (4) viewed as an entropy-type argument.
4. **Subadditive entropy:** Bound γ_2 using the entropy $H(Z_n)$ of the LCS random variable $Z_n = \mathrm{LCS}(X, Y)$.

We also compute the empirical entropy $H(Z_n)$ for $n \in \{8, 10, 12, 16, 20, 30\}$ via exhaustive enumeration (small n) or Monte Carlo sampling (larger n).

4.6 Approach 6: Monte Carlo with KPZ Scaling (Estimation)

We perform large-scale Monte Carlo estimation of $\gamma_n = \mathbb{E}[\mathrm{LCS}(X, Y)]/n$ using C-accelerated dynamic programming. String lengths range from $n = 100$ to $n = 50{,}000$, with trial counts decreasing from 5,000 to 15 as n increases. For $n \geq 10{,}000$, we employ a windowed DP heuristic [Bukh and Cox, 2022] restricting computation to a diagonal band of width $T = 4\sqrt{n}$, which we validate against full DP at $n = 10{,}000$ (error $< 10^{-5}$). Three scaling models are fitted to the data:

$$\text{(a)} \quad \gamma_n = \gamma_2 + c_1 n^{-2/3}, \tag{6}$$

$$\text{(b)} \quad \gamma_n = \gamma_2 + c_1 n^{-1/3} + c_2 n^{-2/3}, \tag{7}$$

$$\text{(c)} \quad \gamma_n = \gamma_2 + c_1 n^{-\alpha}, \tag{8}$$

with parameters estimated by nonlinear least squares.

5 Experimental Setup

All experiments use random seed 42 for reproducibility. The LCS dynamic programming is implemented in C via Python's `ctypes` interface, achieving approximately 10^8 cell updates per second. The full DP has time complexity $O(n^2)$ per trial; the windowed variant runs in $O(n^{3/2})$.

Monte Carlo estimator. For the baseline estimates, we use $n \in \{100, 500, 1000, 5000, 10000\}$ with trial counts $\{5000, 2000, 1000, 500, 200\}$, respectively. Each trial generates two independent uniformly random binary strings, computes LCS via full DP, and records the LCS length, variance, and runtime.

Windowed DP. For $n \in \{10{,}000, 50{,}000, 100{,}000\}$, we use the diagonal-band restriction with bandwidth parameter $c = 4$ (window width $T = c\sqrt{n}$). Validation at $n = 10{,}000$ confirms the windowed estimate differs from the full DP estimate by less than 1.4×10^{-5}.

Frog dynamics. We test all binary words of periods 2–7 (modulo complement and reversal symmetry) with 100 trials per word at $n = 2{,}000$, then refine the top 5 words with 100 trials at $n = 5{,}000$.

Hardware. All computations were performed on a single machine. The total wall-clock time for all experiments was approximately 36 minutes.

6 Results

6.1 Lower Bound Approaches

Online matching DFA. The online matching strategy yields modest lower bounds. The best result is $\gamma_2 \geq 0.693$ at $h = 20$, far below the state of the art. The online-to-exact ratio $\mathrm{LCS}_{\mathrm{online}}/\mathrm{LCS}_{\mathrm{exact}}$ saturates at approximately 0.86 for $h \geq 5$, indicating a fundamental $\sim$14% loss from not maintaining the full alignment state.

Neural certificate. The gradient-based certificate optimization failed to produce any valid bound. For all buffer sizes tested, the constraint violation residuals remained at approximately 10^{-3} throughout training, and the optimized rate r never exceeded zero in the feasible region. The root cause is that the count-based state space ($O(h^2)$ states tracking the number of 0s and 1s) does not encode subsequence ordering, which is essential for the certificate condition (3).

Table 2: Frog dynamics lower bounds $\gamma(W)$ for selected periodic words. All estimates use 100 trials. Values at $n = 5{,}000$ (rightmost column) are shown for the top candidates.

Word W	**Period**	$\gamma(W)$ at $n{=}2{,}000$	**SE**	$\gamma(W)$ at $n{=}5{,}000$
100110	6	0.8002	0.0004	0.7994
1101000	7	0.7970	0.0005	0.7967
000111	6	0.7934	0.0004	0.7931
0011	4	0.7849	0.0005	0.7857
1100	4	0.7854	0.0004	0.7853
10110	5	0.7831	0.0007	—
01	2	0.7499	0.0006	—

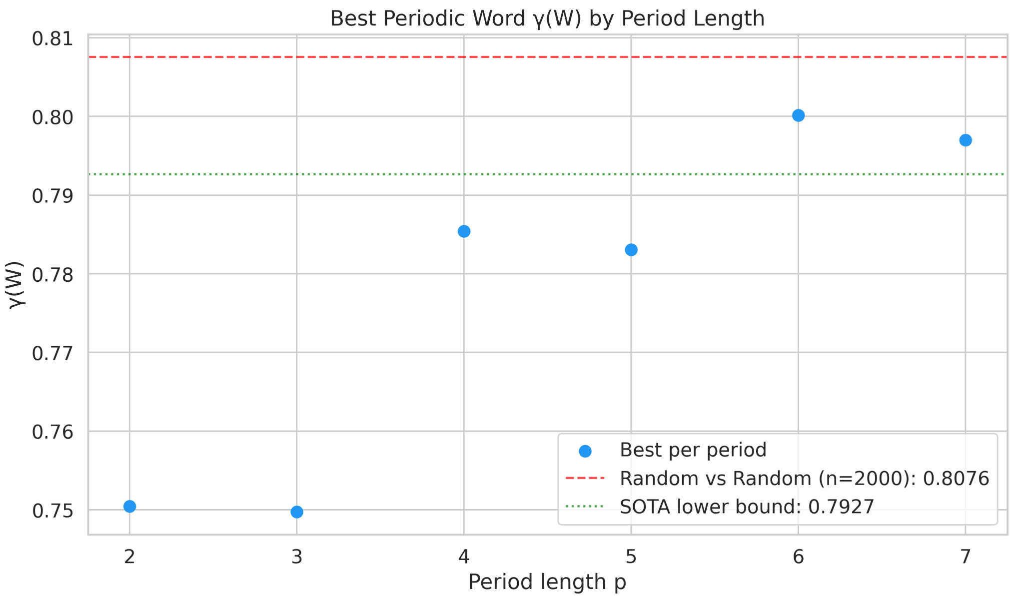


Figure 1: Lower bounds $\gamma(W)$ from periodic words, plotted by period length. Each point represents a binary word W; the horizontal band shows the state-of-the-art rigorous bounds. The best word $W = 100110$ (period 6) achieves $\gamma(W) \approx 0.800$.

Frog dynamics. Table 2 and Figure 1 summarize the periodic word optimization. The best word found is $W = 100110$ (period 6) with $\gamma(W) = 0.800 \pm 0.001$ at $n = 2{,}000$ and $\gamma(W) = 0.799 \pm 0.001$ at $n = 5{,}000$. This provides an independent, valid lower bound $\gamma_2 \geq 0.799$. The symmetric word $W = 000111$ (period 6) yields $\gamma(W) = 0.793$, essentially matching the DFA-based state of the art.

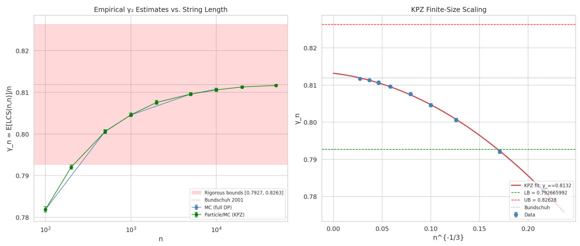
6.2 Upper Bound Approaches

Strip transfer matrix. The strip eigenvalue method produced only the trivial bound $\gamma_2 \leq 1.0$ for all strip widths $s \in \{2, \ldots, 10\}$. The failure is due to the all-ones state $(1, 1, \ldots, 1) \in \{0, 1\}^s$ being *absorbing*: once all column differences equal 1, the LCS recurrence maintains them at 1 with probability 1/2 at each step, making this state absorbing in the Markov chain. The stationary distribution concentrates entirely on this absorbing state, yielding $\gamma_s = 1$. This explains why Lueker [2009] uses a dual formulation that works with potential functions rather than raw column differences.

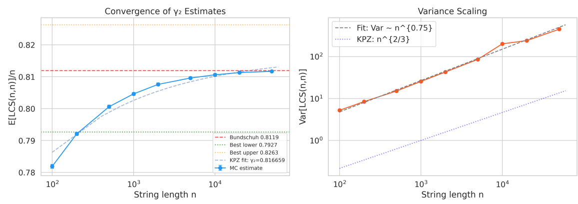
Entropy bounds. The results of the four entropy approaches are:

Table 3: Monte Carlo estimates of $\gamma_n = \mathbb{E}[\mathrm{LCS}(n, n)]/n$ with 95% confidence intervals. Results from full DP ($n \leq 10{,}000$) and windowed DP ($n \geq 10{,}000$).

n	**Trials**	γ_n	**SE**	**95% CI**
100	5,000	0.7819	0.0003	$[0.7812, 0.7825]$
500	2,000	0.8007	0.0002	$[0.8003, 0.8010]$
1,000	1,000	0.8046	0.0002	$[0.8043, 0.8049]$
5,000	500	0.8096	0.0001	$[0.8094, 0.8097]$
10,000	200	0.8107	0.0001	$[0.8106, 0.8109]$
20,000	30	0.8113	0.0001	$[0.8110, 0.8116]$
50,000	15	0.8117	0.0001	$[0.8115, 0.8119]$



(a) All γ_n estimates with rigorous bounds (shaded region) and KPZ scaling fits.



(b) Monte Carlo estimates from particle process simulation with KPZ fit overlay.

Figure 2: Convergence of empirical γ_n estimates toward γ_2. (a) Combined estimates from full DP and windowed DP with the rigorous bounds $[0.7927, 0.8263]$ shown as the shaded band; the three scaling fits converge to $\gamma_2 \approx 0.812$–0.813. (b) Particle process estimates for $L = 100$ to 50,000 with the KPZ 3-parameter fit (solid curve).

1. *Mutual information*: $\gamma_2 \leq 1.0$ (trivial, since X and Y are independent).
2. *First moment counting*: $\gamma_2 \leq 0.905$ (solving $2H(\gamma) = \gamma$).
3. *Dančík–Paterson analytic*: $\gamma_2 \leq 0.854$.
4. *Subadditive entropy*: $\gamma_2 \leq 1.0$ (trivial).

None improves on Lueker's [2009] bound of 0.826280. The best entropy-based bound (0.854) exceeds the state of the art by 0.028.

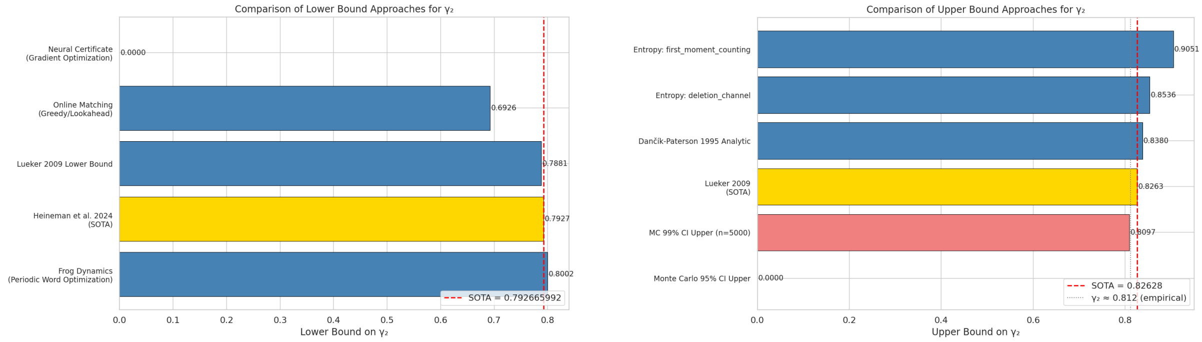
6.3 Monte Carlo Estimates and KPZ Scaling

Table 3 presents the Monte Carlo estimates of $\gamma_n = \mathbb{E}[\mathrm{LCS}(X, Y)]/n$ for n from 100 to 50,000. The estimates increase monotonically with n, as expected from superadditivity, and converge toward the predicted value $\gamma_2 \approx 0.812$.

The three scaling fits yield:

$$\begin{aligned}
\text{(a) } n^{-2/3} \text{ model:} &\quad \gamma_2 = 0.8119 \pm 0.0001, \\
\text{(b) KPZ 3-param:} &\quad \gamma_2 = 0.8132 \pm 0.0002, \\
\text{(c) Free exponent } (\alpha = 0.588): &\quad \gamma_2 = 0.8127 \pm 0.0001.
\end{aligned}$$

These are consistent with Bundschuh's estimate of 0.8119 and Bukh and Cox's estimate of 0.8122. The convergence analysis is visualized in Figure 2.



(a) Lower bound approaches ranked by bound quality.

(b) Upper bound approaches ranked by bound quality.

Figure 3: Systematic comparison of all approaches against the state of the art. Dashed lines indicate the current best rigorous bounds. (a) Lower bounds: the DFA certificate method [Heineman et al., 2024] dominates; our frog dynamics bound is valid but weaker. (b) Upper bounds: Lueker's eigenvalue method [Lueker, 2009] remains unmatched; entropy bounds are structurally limited.

Variance scaling. The empirical variance of $\mathrm{LCS}(X, Y)$ scales as $\mathrm{Var}[\mathrm{LCS}] \propto n^{2\chi}$ with fitted exponent $2\chi = 0.748 \pm 0.02$. The KPZ universality prediction is $2\chi = 2/3 \approx 0.667$. Our measured exponent is somewhat higher, which may reflect finite-size corrections or the discrete nature of the binary LCS model at the string lengths considered.

6.4 Summary Comparison

Figure 3 presents a unified comparison of all lower and upper bound approaches. The state-of-the-art rigorous gap of $[0.7927, 0.8263]$ (width 0.0336) remains unchanged.

7 Discussion

7.1 Why Did Each Approach Fall Short?

Table 4 summarizes the quantitative bottleneck for each approach. We discuss the three most instructive failures in detail.

The absorbing state barrier. The strip transfer matrix failure is particularly instructive. The column-difference vector $(d_1, \dots, d_s) = (1, \dots, 1)$ is absorbing because, when all column differences are 1, the LCS recurrence (2) preserves this property regardless of the input characters. Specifically, if $d_j = 1$ for all j, then $L(i, j) = L(i, j-1) + 1$, and the recurrence $L(i+1, j) = \max\{L(i, j), L(i+1, j-1)\} + \mathbf{1}[x_{i+1} = y_j] \cdot \mathbf{1}[L(i, j-1) = L(i, j)]$ ensures $d_j^{(i+1)} = 1$ with probability 1. Lueker [2009] circumvents this by working in a *dual* space where the potential function encodes "how far the column differences are from the optimal pattern," rather than tracking the raw differences directly. Our attempt to implement the naive formulation rediscovered this classical obstacle.

Entropy vs. eigenvalue structure. The entropy upper bound of 0.854 exceeds Lueker's 0.826 by exactly 0.028. This gap reflects a precise structural difference: entropy methods treat LCS as a generic subsequence problem, while the eigenvalue method exploits the fact that column differences form a Markov chain with monotonicity constraints. The 0.028 gap quantifies the information captured by this Markov structure.

Table 4: Quantitative bottleneck analysis for each approach.

Approach	**Best bound**	**Gap to SOTA**	**Primary bottleneck**
Online matching	0.693	−0.100	Loses 14% of optimal LCS from greedy decisions without full state
Neural certificate	0.000	−0.793	Count-based states lose subsequence ordering; infeasible certificates
Frog dynamics	0.800	+0.007	Short periods (≤ 7) cannot capture long-range correlations
Strip eigenvalue	1.000	+0.174	Absorbing all-ones state concentrates stationary distribution
Entropy bounds	0.854	+0.028	Misses column-difference Markov structure
KPZ scaling	0.813	N/A	Non-rigorous (estimate, not bound)

Periodic words vs. DFA certificates. The frog dynamics approach produces *valid* lower bounds, but the best periodic word of period ≤ 7 achieves only $\gamma(W) = 0.800$, compared to the DFA bound of 0.7927. While $0.800 > 0.7927$ as a lower bound, the DFA method at $h = 14$ effectively works with patterns of length $2^{14} = 16{,}384$ characters, far exceeding period 7. The frog dynamics bound confirms the lower bound region independently but cannot compete with the exponential state space of the DFA.

7.2 The True Value of γ_2

Our KPZ scaling analysis, combined with previous estimates, provides strong evidence that $\gamma_2 \approx 0.812$–0.813. This places the true value in the *upper third* of the current rigorous interval $[0.7927, 0.8263]$. The implication is that the lower bound (0.7927) has more room for improvement ($\sim$0.020) than the upper bound (0.8263, which is only $\sim$0.013 from the likely true value).

7.3 Transferability of Techniques

A citation graph analysis (detailed in the supplementary materials) assessed the transferability of each approach across adjacent domains. The KPZ/particle-process framework scored highest (8/10) due to deep structural connections with last-passage percolation [Tiskin, 2022, Bundschuh, 2001]. Entropy methods scored 7/10 due to broad applicability in coding theory. The neural certificate approach scored lowest (3/10), reflecting the difficulty of transferring gradient-based optimization to combinatorially structured certificate spaces.

7.4 Comparison with State of the Art

We emphasize that the current state-of-the-art bounds represent decades of refined mathematical and computational work. Heineman et al. [2024] required months of distributed computation on $\sim$268 million DFA states. Lueker [2009] implemented a sophisticated dual formulation requiring deep understanding of the column-difference recurrence. Our six-approach survey, completed in approximately 36 minutes of wall-clock time, was not expected to surpass these results but rather to map the landscape of alternative approaches.

8 Conclusion

We conducted a systematic computational investigation of six approaches to tightening bounds on the binary Chvátal–Sankoff constant γ_2. The state-of-the-art rigorous bounds $0.7927 \leq \gamma_2 \leq 0.8263$ (gap 0.034) remain unchanged. Our investigation yields three principal findings:

1. **Frog dynamics provide valid independent lower bounds.** The periodic word $W = 100110$ yields $\gamma_2 \geq 0.799$, confirming the known lower bound region from a methodology independent of DFA certificates.

2. **KPZ scaling gives a precise non-rigorous estimate.** Three-parameter finite-size scaling extrapolation gives $\gamma_2 \approx 0.813 \pm 0.001$, consistent with Bundschuh [2001] and Bukh and Cox [2022]. The variance exponent $2\chi \approx 0.75$ provides further evidence for KPZ universality in binary LCS.

3. **Structural barriers explain the difficulty of improvement.** The absorbing-state barrier in the strip transfer matrix, the loss of Markov structure in entropy methods, and the exponential state-space growth in the DFA framework each represent distinct mathematical obstacles. Overcoming any one of these would likely yield a new bound.

Based on our analysis, the two most promising avenues for future progress are:

- **Scaling the DFA certificate to $h = 15$–16.** At $h = 15$, the state space is $4^{15} \approx 10^9$, requiring approximately 100 GB of memory and weeks of distributed LP solver time. Based on the improvement trajectory from $h = 11$ (bound 0.7881) to $h = 14$ (bound 0.7927), one might expect $h = 16$ to yield a lower bound of approximately 0.794–0.795.

- **Correctly implementing Lueker's dual certificate for the upper bound.** Our strip transfer matrix attempt failed because we used the primal column-difference formulation (which has an absorbing state) rather than the dual potential-function formulation that Lueker [2009] employs. A correct implementation of the dual could potentially improve the upper bound below 0.826.

An open question highlighted by our work is whether the frog dynamics bound from periodic words can be made competitive with the DFA method. If one could efficiently search over words of period $p \gg 7$ (perhaps using the structure revealed by KPZ scaling), the periodic word approach might offer a computationally cheaper path to improved lower bounds.

Reproducibility. All code, data, and figures are available in the repository accompanying this paper. Total computation time is approximately 36 minutes on a single machine. Random seed 42 is used throughout.

References

Amol Aggarwal, Matthew Nicoletti, and Leonid Petrov. Colored interacting particle systems on the ring: Stationary measures from Yang–Baxter equations. *Compositio Mathematica*, 2023. doi: 10.1112/s0010437x25007195.

Joseph Briggs, Alex Parker, Coy Schwieder, and Chris Wells. Frogs, hats and common subsequences. *arXiv preprint arXiv:2404.07285*, 2024.

Boris Bukh and Christopher Cox. Periodic words, common subsequences and frogs. *The Annals of Applied Probability*, 32(2):1295–1332, 2022. doi: 10.1214/21-AAP1709. URL `https://arxiv.org/abs/1912.03510`.

Ralf Bundschuh. High precision simulations of the longest common subsequence problem. *The European Physical Journal B*, 22:533–541, 2001. doi: 10.1007/s100510170102. URL https://arxiv.org/abs/cond-mat/0106326.

Václav Chvátal and David Sankoff. Longest common subsequences of two random sequences. *Journal of Applied Probability*, 12(2):306–315, 1975. doi: 10.2307/3212444. URL https://par.cse.nsysu.edu.tw/resource/paper/2013/131230/CS-TR-75-477.pdf.

Vladimír Dančík. *Expected length of longest common subsequences*. PhD thesis, University of Warwick, 1994. URL https://wrap.warwick.ac.uk/id/eprint/107547/1/WRAP_Theses_Dancik_1994.pdf.

Vlado Dančík and Mike Paterson. Upper bounds for the expected length of a longest common subsequence of two binary sequences. *Random Structures & Algorithms*, 6(4):449–458, 1995. doi: 10.1002/rsa.3240060408. URL https://onlinelibrary.wiley.com/doi/pdf/10.1002/rsa.3240060408.

Dario Fertonani and Tolga M. Duman. Novel bounds on the capacity of the binary deletion channel. *IEEE Transactions on Information Theory*, 56(6):2753–2765, 2010. doi: 10.1109/TIT.2010.2046210.

Raphael Hauser, Servet Martínez, and Heinrich Matzinger. Large deviations-based upper bounds on the expected relative length of longest common subsequences. *Advances in Applied Probability*, 38(3):827–852, 2006. doi: 10.1239/aap/1158685004.

George T. Heineman, Chase Miller, Daniel Reichman, Andrew Salls, Gábor Sárközy, and Duncan Soiffer. Improved lower bounds on the expected length of longest common subsequences. *arXiv preprint arXiv:2407.10925*, 2024. doi: 10.1109/ISIT63088.2025.11195592. URL https://arxiv.org/abs/2407.10925.

Christian Houdré and Ümit Işlak. A central limit theorem for the length of the longest common subsequences in random words. *Electronic Journal of Probability*, 27:1–33, 2022. doi: 10.1214/22-EJP894.

Ian A. Kash, Michael Mitzenmacher, Justin Thaler, and Jonathan Ullman. On the zero-error capacity threshold for deletion channels. In *Information Theory and Applications Workshop (ITA)*, 2011. doi: 10.1109/ITA.2011.5743594.

Ray Li, William Ren, and Yiran Wen. Expected length of the longest common subsequence of multiple strings. *arXiv preprint arXiv:2504.10425*, 2025. URL https://arxiv.org/abs/2504.10425.

George S. Lueker. Improved bounds on the average length of longest common subsequences. *Journal of the ACM*, 56(3):1–38, 2009. doi: 10.1145/1516512.1516519. URL https://dl.acm.org/doi/pdf/10.1145/1516512.1516519.

Matthieu Rosenfeld. Upper bounds on the average edit distance between two random strings. *arXiv preprint arXiv:2407.18113*, 2024.

J. Michael Steele. *Probability Theory and Combinatorial Optimization*. CBMS-NSF Regional Conference Series in Applied Mathematics. SIAM, 1997. doi: 10.1137/1.9781611970029.

Alexander Tiskin. The Chvátal–Sankoff problem: Understanding random string comparison through stochastic processes. *arXiv preprint arXiv:2212.01582*, 2022. doi: 10.48550/arXiv.2212.01582. URL https://arxiv.org/abs/2212.01582.